

As President of SEDAR Tug Services, Inc., I firmly believe that the continued growth and competitiveness of our company require us to embrace digital transformation. The complexity of managing tug operations can no longer rely on disconnected systems, manual processes, or separate Excel files maintained by individual departments.

Our vision is to implement a fully integrated **Enterprise Resource Planning (ERP)** and **Marine Fleet Management System** that will unify all core business functions into a single platform. This system will seamlessly connect Finance, Tug Operations, Technical Maintenance, Crewing, Health, Safety, Security and Environment (HSSE), Procurement, Inventory Management, Human Resources, Marketing, and Executive Management.

With a centralized system, information will be entered once and shared across the organization in real time, providing a single source of truth for decision-making. This will eliminate duplicate data entry, improve operational efficiency, strengthen internal controls, enhance regulatory compliance, and provide management with accurate, timely, and reliable information.

Our objective is not simply to automate existing processes but to build a modern, data-driven organization capable of supporting sustainable growth, delivering world-class marine services, and meeting the evolving needs of our clients and stakeholders. Through this transformation, SEDAR Tug Services, Inc. will establish itself as one of the most professionally managed and technologically advanced tug service providers in the Philippines.

This initiative represents a strategic investment in the future of our company—one that will strengthen operational excellence, improve financial transparency, optimize asset utilization, enhance customer satisfaction, and reinforce our commitment to safety, reliability, and corporate governance.

The ideal system should include these modules:

Department Required System Finance & Accounting General Ledger, Accounts Payable, Accounts Receivable, Budgeting, Cash Flow, Fixed Assets, Payroll Tug Operations Job dispatch, tug scheduling, voyage logs, fuel monitoring, towage billing, GPS/AIS tracking Technical/Maintenance Planned Maintenance System (PMS), dry docking, work orders, spare parts, equipment history HSE Incident reporting, inspections, permits, audits, risk assessments, training records Crewing Crew scheduling, certifications, STCW documents, medicals, leave, payroll integration Procurement Purchase Requests, Purchase Orders, supplier management, approval work flow Inventory Spare parts, fuel, lubricants, office supplies, warehouse management, barcode support HR Employee records, attendance, performance evaluation, recruitment Document Control Contracts, vessel certificates, insurance, permits, board resolutions, ISO documents Management Dashboard KPIs, financial reports, utilization, vessel availability, profitability

Recommended Systems

1. Microsoft Dynamics 365 Business Central

Best for medium-sized companies:

Excellent accounting

Purchasing

Inventory

HR integration

Power BI dashboards

Can be customized for tug operations

2. Odoo (All-in-one ERP platform) / OnDemand (SaaS) Offer from OpenERP

One of the best values. Modules include:

Accounting

Fleet

Maintenance

Inventory

Purchase

HR

Payroll

CRM

Documents

Mobile app

It is highly customizable and considerably less expensive than many enterprise systems.

3. Star Information Systems Ship Net

Designed specifically for shipping companies. Includes:

Technical management

Planned maintenance

Procurement

Inventory

HSEQ

Crewing

Finance

4. ABS NS5

Widely used by commercial shipping operators. Excellent for:

Maintenance

HSE

Purchasing

Crewing

Compliance

Other Essential Programs

Power BI for executive dashboards and KPI reporting.

Microsoft 365 for email, document collaboration, Teams, and SharePoint.

DocuSign for digital approvals and contracts.

AIS/GPS fleet tracking software for real-time vessel monitoring.

Suggested System Architecture for SEDAR

Finance & Accounting

General Ledger

Budgeting

Cash Management

Payroll

Bank Reconciliation

Marine Operations

Dispatch

Tug Scheduling

Job Orders

Fuel Monitoring

Customer Billing

Technical Department

Planned Maintenance

Dry Dock Planning

Defect Reporting

Spare Parts

Work Orders

Crewing

Crew Rotation

STCW Expiry Tracking

Medical Certificates

Payroll

Training

Safety/HSE

Incident Reporting

Near Miss Reports

Risk Assessments

Safety Meetings

Compliance Audits

Inventory

Spare Parts

Fuel
Lubricants
Office Supplies
Barcode Tracking
Corporate Management
Board Resolutions
Legal Cases
Insurance
Contracts
Internal Audit
KPI Dashboard
My recommendation for SEDAR

Given SEDAR's size and our plans to strengthen corporate governance and internal controls, I would recommend:

Odoo as the core ERP because it provides accounting, procurement, inventory, HR, maintenance, and document management at a reasonable cost and can be customized for tug operations.

Add a dedicated maritime maintenance and fleet module if your fleet grows or maintenance needs become more complex.

Use Power BI for executive dashboards so the Board and CEO can monitor financial performance, fleet utilization, maintenance status, HSE metrics, and profitability in real time.

This combination offers a practical balance between capability, scalability, and cost for a tugboat operator like SEDAR.